

# OpenERP User Guide

## Property Management System

Reinaldo Crespo

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# Chapter 1

## OpenERP User's Guide

**Audience:** End users, system administrators, and business analysts operating an OpenERP installation. For installation and server setup, see [INSTALL.md](#).

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### 1.1 Table of Contents

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### 1.2 1. Getting Started

#### 1.2.1 1.1 First Login

Navigate to `http://<server>:5173` in a browser. Log in with the credentials provided by your administrator. The default admin account created at install time is `admin / admin` — **change this password immediately** under Admin → Users before going live.

The session token expires after 8 hours. You will be redirected to the login screen automatically when it expires.

#### 1.2.2 1.2 Dashboard

The Dashboard shows live KPIs pulled from the database:

| Tile       | What it shows                                                        |
|------------|----------------------------------------------------------------------|
| Cash       | Balance of the GL account configured as <code>gl_cash_account</code> |
| AR Balance | Sum of open AR invoice balances                                      |
| AP Balance | Sum of open AP bill balances                                         |
| Open SO    | Count and value of sales orders in OPEN or PARTIAL status            |
| Open PO    | Count and value of purchase orders in OPEN or PARTIAL status         |
| Overdue AR | Invoices past due date                                               |

Check the Dashboard every morning. A rising AR Balance combined with a flat Cash balance is an early warning that collections need attention. A rising AP Balance approaching its due dates means a check run is due.

### 1.2.3 1.3 Company Setup

Go to **Admin → Company Settings** to configure:

- **Company name, address, phone, website** — printed on all PDFs
- **Logo** — upload a PNG or JPG; embedded in invoices, statements, SOs, and POs
- **GL account mappings** — tells the system which GL account to debit/credit for AR, AP, cash, revenue, COGS, inventory, and tax payable. These are the `gl_*_account` config keys. Without them the system falls back to the first active account with the matching subtype.
- **PDF document templates** — configurable header subtitle, footer note, and per-document terms text (invoice terms, SO terms, PO terms)
- **SMTP** — for emailing invoices, SOs, and POs directly from the system

**Do this first.** Until the GL account mappings are configured, the auto-posting engine posts to the first matching account it finds. If your chart of accounts has multiple accounts of the same subtype, the wrong account may receive the entry. Set the mappings before processing any transactions.

### 1.2.4 1.4 Chart of Accounts

Before processing any transactions, import or manually create your chart of accounts under **GL → Accounts**.

Each account has: - **Account number** — user-defined (e.g. 1000, 4000-01) - **Account name** - **Account type** — ASSET, LIABILITY, EQUITY, REVENUE, EXPENSE - **Account subtype** — AR, AP, CASH, SALES, COGS, INVENTORY, etc. The subtype is used by the auto-posting engine to resolve accounts from the config keys above. - **Is header** — header accounts cannot receive journal line amounts; they exist only for subtotalling in financial statements.

Use **GL → Fiscal Years** to create the current fiscal year, then **GL → Periods** to create the twelve monthly periods inside it. The system will not allow posting to a closed period.

### 1.2.5 1.5 Payment Terms and Tax Codes

- **Admin → Payment Terms** — define NET30, NET60, 2/10-NET30, etc. Used on customer and vendor records and defaulted onto SOs, POs, invoices, and bills.
- **Admin → Tax Codes** — define tax rates (e.g. TX-STATE at 6.25%). Assigned to items and overridable per invoice line. For US sales tax with multi-jurisdiction support, use the Sales Tax engine (Admin → Sales Tax).

## 1.2.6 1.6 Go-Live Setup Checklist

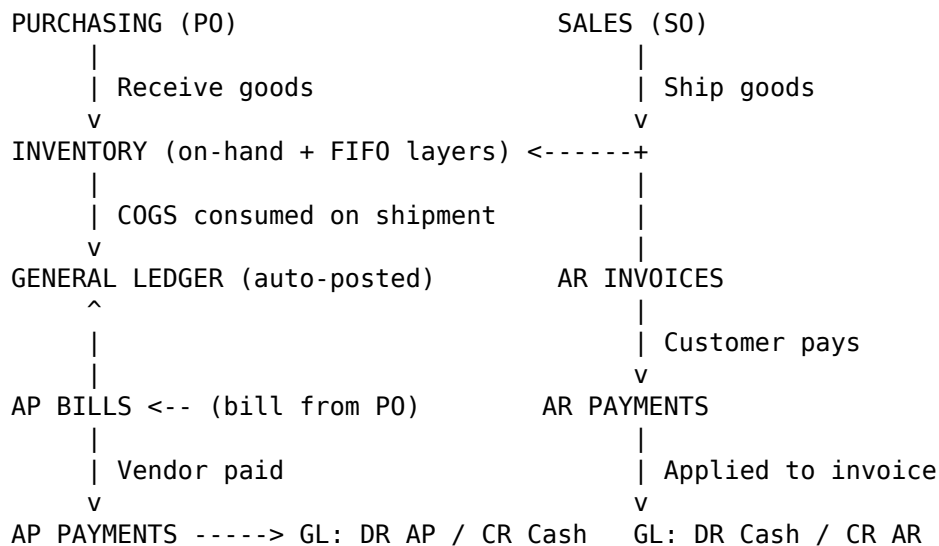
Before your first live transaction, confirm the following:

- ☐ Company name, address, and logo are entered in Company Settings
- ☐ GL account mappings (all gl\_\*\_account keys) are set
- ☐ Fiscal year and all 12 periods are created
- ☐ Chart of accounts is imported or built
- ☐ Payment terms are created and assigned to customers and vendors
- ☐ Tax codes or sales tax jurisdictions are configured
- ☐ Item master is loaded (items, categories, UOM, price lists)
- ☐ Customer master is loaded with credit limits and terms
- ☐ Vendor master is loaded with 1099 flags and terms
- ☐ Admin user password changed from default
- ☐ Non-admin user accounts and groups are created
- ☐ Warehouse and bin locations are created
- ☐ Opening inventory balances are entered via manual adjustments
- ☐ Opening AR balances are entered as invoices
- ☐ Opening AP balances are entered as bills
- ☐ Opening GL balances are imported (Admin → Import → GL Balances)

## 1.3 2. How the Modules Connect

Understanding how data flows between modules prevents duplicate work and explains why a transaction in one area shows up somewhere unexpected.

### 1.3.1 2.1 The Core Flows



Every transaction that touches money posts a GL batch automatically. You do not need to create manual journal entries for normal operations.

### 1.3.2 2.2 Auto-Posting: What Posts to GL and When

The following events create a GL journal batch silently in the background the moment you save:

| Event                               | GL Debit  | GL Credit                                        |
|-------------------------------------|-----------|--------------------------------------------------|
| AR Invoice saved                    | AR        | Revenue (+ Tax Payable)                          |
| AR Payment applied to invoice       | Cash      | AR                                               |
| AP Bill saved                       | Expense   | AP                                               |
| AP Payment applied to bill          | AP        | Cash                                             |
| SO Shipment posted                  | COGS      | Inventory                                        |
| Inventory Adjustment (positive qty) | Inventory | Inv-Adj                                          |
| Inventory Adjustment (negative qty) | Inv-Adj   | Inventory                                        |
| AR Credit Memo created              | Revenue   | AR                                               |
| AP Debit Memo created               | AP        | Expense                                          |
| PO Receive (no direct GL)           | —         | — (FIFO layer created; GL posts when items ship) |

**Key implication:** When you receive a PO, the goods enter inventory and a FIFO cost layer is created, but no GL entry posts at that moment. The GL entry (COGS) posts when those items are later shipped on a Sales Order. This is the standard perpetual inventory model — the inventory asset account increases on receipt via the AP bill (DR Inventory is embedded in the COGS flow), and COGS hits the income statement on shipment.

### 1.3.3 2.3 The AR-to-Cash Chain

Many users expect cash to appear in their bank account when they “apply” a payment. What actually happens is a GL entry (DR Cash / CR AR). The cash account balance rises, the AR balance falls. The actual bank deposit must be reconciled in **Bank Reconciliation** (\$7.8) when the bank statement confirms the funds cleared.

|                    |                                                     |
|--------------------|-----------------------------------------------------|
| Invoice saved      | → AR balance rises, Revenue balance rises           |
| Payment saved      | → No effect yet (payment is unmatched)              |
| Payment applied    | → AR balance falls, Cash (GL) balance rises         |
| Bank statement rec | → Statement line matched to payment, bank confirmed |

### 1.3.4 2.4 The PO-to-Expense Chain

|                    |                                                          |
|--------------------|----------------------------------------------------------|
| PO created         | → No GL effect                                           |
| PO received        | → On-hand rises; FIFO layer created                      |
| AP Bill from PO    | → AP balance rises, Expense (or Inventory) balance rises |
| AP Payment applied | → AP balance falls, Cash balance falls                   |

For **inventory items**: the AP bill posts DR Inventory / CR AP — the cost stays in inventory until the items ship, at which point COGS is posted. For **service or expense items** (non-inventory): the AP bill posts DR Expense / CR AP directly.



### 1.3.5 2.5 How a Sales Order Touches Five Modules

When a Sales Order is created, processed, and invoiced:

1. **SO created** — no financial effect; just a commitment
2. **SO shipped** — inventory on-hand falls; FIFO cost layers consumed; GL: DR COGS / CR Inventory
3. **SO invoiced → AR Invoice** — AR balance rises; GL: DR AR / CR Revenue
4. **AR Payment applied** — AR balance falls; GL: DR Cash / CR AR
5. **Bank Reconciliation** — payment matched to bank deposit; GL cash confirmed

The sales rep's commission accrues conceptually at step 3 (when the invoice is created). The Commission Report at **Sales → Commissions** shows commission earned on invoiced SOs.

### 1.3.6 2.6 ATP (Available to Promise) and How Inventory Informs Sales

When you select an item on a Sales Order, the **ATP badge** shows:

ATP = On-hand – Committed (open SO lines) + On-order (open PO lines)

A green badge means you can promise this quantity to the customer. A red badge means either you will need to back-order or you need to place a PO first. The ATP calculation happens in real time and reflects all open SOs and POs — it is the link between sales commitments and purchasing plans.

## 1.4 3. Order-to-Cash

The Order-to-Cash workflow moves a sales opportunity from quotation through cash collection.

Customer → Quote → Sales Order → Shipment → AR Invoice → Cash Receipt → Applied → Reconciled

### 1.4.1 3.1 Customer Master

**AR → Customers** — create a customer before entering any orders.

Key fields: - **Customer number** — unique identifier; printed on documents - **Credit limit** — if set, the system blocks SO creation when the customer's open AR + open SO value exceeds this limit - **On credit hold** — manually block all new orders for this customer; a red banner appears on the SO form when a held customer is selected - **Default sales rep** — auto-populated on new SOs when the customer is chosen - **Customer class** — used by matrix pricing (e.g. KEY, STANDARD, PROSPECT) - **Payment terms** — defaulted onto invoices

**Ship-to addresses:** Click **Ship-To** on the customer row to manage multiple delivery addresses. One can be flagged as default; it auto-selects on the SO form.

**Contacts:** Click **Contacts** to add named contacts with phone and email. Contacts flagged "primary" receive emailed documents.

**What to do next after adding a customer:** Set the credit limit and payment terms before their first order. A customer with no credit limit will never be blocked by the credit check, regardless of how much they owe.

### 1.4.2 3.2 Sales Quotes

**Sales → Quotes** — optional step before creating a firm order.

Statuses: DRAFT → SENT → ACCEPTED or EXPIRED or CANCELLED

- **Send** converts a DRAFT quote to SENT (recordkeeping only; no email is sent automatically unless configured).
- **Accept** creates a Sales Order from the quote lines and marks the quote ACCEPTED.
- **Expire** marks the quote EXPIRED; no order is created.

**What comes next:** After a customer accepts a quote (verbally or in writing), click **Accept** to convert it to a Sales Order. The quote line prices and quantities carry over exactly. You can then edit the SO if quantities need adjustment before shipment.

### 1.4.3 3.3 Sales Orders

**Sales → Orders** — the core transaction.

#### 1.4.3.1 Creating a Sales Order

1. Choose a **customer**. The system immediately checks credit: a green/yellow/red banner shows available credit. If the customer is on hold, the **Create Order** button is disabled.
2. Set **order date**, **requested ship date** (optional), **period** (accounting period for GL posting), **payment terms**, **ship-to address**, and **sales rep**.
3. Add **line items**. For each line:
  - Select an **item**. The system fetches the price (checking contract → customer price list → matrix pricing → base price list → list price, in that order) and shows the **ATP** (available-to-promise) badge in green or red.
  - Enter **qty ordered**. If qty breaks apply the price may adjust.
  - Optionally set a **promise date** for this specific line.
  - Check **drop-ship** if the vendor should ship directly to the customer (see §3.7).
4. Save. The order is created in OPEN status.

**What comes next after saving a Sales Order:** - If ATP is red on any line, go to **Purchasing → Orders** and place a PO for the shortfall. Or go to **Inventory → Reorder Suggestions** to see all understocked items at once. - If the order should ship today, proceed to the **Ship** action (below). - If payment terms are prepaid, go to **AR → Deposits** to record the customer's payment before shipping.

#### 1.4.3.2 Sales Order Statuses

| Status    | Meaning                  | What to do                                  |
|-----------|--------------------------|---------------------------------------------|
| OPEN      | No lines shipped yet     | Pick and pack; click Ship when ready        |
| PARTIAL   | Some lines shipped       | Ship remaining lines or create a back-order |
| SHIPPED   | All lines shipped        | Click Invoice to generate the AR invoice    |
| INVOICED  | AR invoice generated     | Monitor payment in AR → Payments            |
| CANCELLED | Voided before fulfilment | No further action needed                    |

### 1.4.3.3 Shipping

Click **Ship** on an order in OPEN or PARTIAL status. Enter: - **Ship date, period, warehouse** - Per-line **qty shipped** (defaults to remaining qty) - Optionally: BOL number, carrier pro number, pieces, weight, freight class

The system: - Creates SHIPMENT inventory transactions (reduces on-hand) - Consumes FIFO cost layers and calculates COGS - Posts a GL batch: DR COGS / CR Inventory - Updates line statuses; sets header to PARTIAL or SHIPPED

**After shipping:** Print or email the packing list (PDF button). The order status moves to SHIPPED. If not all lines shipped, status is PARTIAL — you can ship again or create a back-order. When all lines are SHIPPED, the **Invoice** button appears.

### 1.4.3.4 Invoicing

Once fully shipped (SHIPPED status), click **Invoice**. Provide an invoice number, invoice date, due date, and period. The system creates an AR invoice from the SO lines with the exact shipped quantities.

**After invoicing:** The AR Invoice is created automatically and GL auto-posts (DR AR / CR Revenue). Email the invoice to the customer's primary contact using the **Email** button on the invoice. Then monitor payment from **AR → Payments**. If the customer does not pay by the due date, the invoice will appear in **AR → Aging** under the overdue buckets.

### 1.4.3.5 Back-orders

Click **Back-Order** on a PARTIAL order to split remaining (unshipped) qty into a new linked SO. The original order's lines are reduced to shipped qty; the new order covers the remainder.

**When to use back-orders vs waiting:** If you expect stock within a few days, leaving the order PARTIAL is fine. If the customer needs their invoice now and stock will take weeks, create a back-order so you can invoice the shipped portion immediately. The back-order report (**Sales → Back-Orders**) shows all open back-orders so nothing falls through the cracks.

### 1.4.3.6 Drop-Ship Orders

After adding drop-ship lines to an SO, click **Drop-Ship PO**. Choose a vendor, PO number, dates, period, and terms. The system creates a PO with the customer's shipping address; when the vendor ships and the PO is received, the linked SO lines are marked shipped automatically — no warehouse inventory movement occurs.

## 1.4.4 3.4 AR Invoices

**AR → Invoices** — invoices can be created standalone or from an SO (§3.3).

Standalone invoice creation: 1. Choose customer, invoice number, dates, period, payment terms. 2. Add lines: item, description, qty, unit price, tax rate, and optionally a jurisdiction override. 3. The system computes subtotal, tax, and total. On save it posts a GL batch: DR AR / CR Revenue (+ CR Tax Payable if tax > 0).

### 1.4.4.1 Voiding an Invoice

Only OPEN invoices can be voided. Click **Void**. The system posts a reversing GL batch.

**When to void vs write-off:** Void an invoice when it was created in error (wrong customer, wrong amount, duplicate). Write-off an invoice when the goods were delivered, the invoice is legitimate, but the customer will not pay. A void removes the revenue from your books; a write-off moves the balance to Bad Debt Expense.

#### 1.4.4.2 Write-offs

For uncollectable balances, click **Write-Off** on an OPEN or PARTIAL invoice. Enter the write-off amount. The system creates an ar\_write\_offs record and posts GL: DR Bad Debt Expense / CR AR.

#### 1.4.5 3.5 Cash Receipts

**AR → Payments** — record a cash receipt: 1. Choose customer, payment number, date, amount, payment method (CHECK, WIRE, ACH, etc.), and period. 2. Save. The payment is unmatched (status: OPEN). 3. Click **Apply**. Select open invoices and enter the amount to apply to each. The system updates invoice balances and posts GL: DR Cash / CR AR.

**What comes next after applying a payment:** - The invoice balance falls (possibly to \$0, which marks it PAID). - The Cash GL account rises. - The physical deposit must still be matched to your bank statement in **Bank Reconciliation** (§7.8) when the statement is available. - If the customer took an early-pay discount (e.g., 2/10-NET30) and paid less, apply their actual amount and enter the discount on the apply screen. The remaining balance is posted as discount taken.

##### 1.4.5.1 Credit Memos

**AR → Credit Memos** — issue a credit for a return or price adjustment: 1. Create with customer, amount, and description. Posts GL: DR Revenue / CR AR. 2. Apply to an open invoice using **Apply** — reduces the invoice balance. 3. **Void** a credit memo to reverse it (GL reversal posted).

**Credit memo vs voiding the invoice:** If the customer received goods and is returning some of them, issue a credit memo — the original invoice was correct when issued. If the invoice itself was wrong (wrong price, wrong customer), void the invoice and reissue.

#### 1.4.6 3.6 Customer Deposits

**AR → Deposits** — record a prepayment before the invoice is raised: 1. Create deposit: customer, amount, period. 2. Apply to an invoice when it is ready: GL moves the deposit from Deposits liability to AR. 3. **Void** reverses the application or the original deposit.

**Procedure for prepaid customers:** Record the deposit as soon as you receive the check or wire. Ship and invoice as normal. When applying the deposit to the invoice, the invoice balance will drop to zero (or to the remaining amount if the deposit was partial). This keeps your AR aging accurate — the deposit does not show as an open invoice.

#### 1.4.7 3.7 AR Aging and Collections

**AR → Aging** — shows open balances bucketed by 0-30, 31-60, 61-90, 91+ days past due.

**How to run collections from the Aging screen:** 1. Run the Aging report weekly (or daily for high-volume operations). 2. Focus the 61-90 and 91+ columns first

— these are the most at risk. 3. For each overdue customer, click through to the customer record to see the contact name and phone. 4. Log your contact attempt in **AR → Collections** with notes. 5. Customers who are 90+ days with no response: discuss write-off with management. 6. Customers who promise to pay: set a follow-up date in the Collections log.

**AR → Collections** — a prioritized queue of customers with overdue balances. Log contact attempts with notes; the contact history is visible per customer.

### 1.4.8 3.8 Customer Statements

**AR → Statements** — view a customer's transaction history for any date range. **Print** generates a PDF. **Email** sends it to the customer's primary contact.

**When to send statements:** Send monthly statements to all customers with open balances on the last business day of the month. Many customers use the statement (not individual invoices) to schedule their payments. A statement shows all open invoices, any credits, and the total balance due.

## 1.5 4. Procure-to-Pay

Vendor → Requisition → RFQ → PO → Receipt → AP Bill → Payment → Applied → Reconciled

### 1.5.1 4.1 Vendor Master

**AP → Vendors** — create vendors before entering bills or POs.

Key fields: - **Tax ID + Is 1099** — populated on the 1099 report at year end - **Currency** — if not USD, bills are entered in the vendor's currency; FX gain/loss is computed when payment is applied - **Payment terms** — defaulted onto bills

**Vendor bank accounts** — under **AP → Vendor Bank Accounts**; used for ACH/EFT export.

**What to do next after adding a vendor:** Set payment terms and, if applicable, the currency. If the vendor is a US sole proprietor or LLC who provides services (not goods), check the **Is 1099** box and enter their Tax ID (EIN or SSN) — you will need this at year end. Set a default expense GL account if all purchases from this vendor hit the same expense line.

### 1.5.2 4.2 Purchase Requisitions

**Purchasing → Requisitions** — internal request before a PO is issued.

Statuses: DRAFT → PENDING → APPROVED / REJECTED → CONVERTED or CANCELLED

- **Submit** sends a DRAFT to PENDING for approval.
- **Approve** (requires system.users.manage or an approver role) advances to APPROVED and stamps approved\_by.
- **Convert** turns an APPROVED requisition into a PO; choose vendor, PO number, dates, period, and terms in the modal.

**When to use requisitions:** In organizations where purchasing approval is required before committing spend, requisitions are the gate. The requesting employee creates a DRAFT, submits it, and a manager approves or rejects. Only after

approval can a PO be created. For small purchases below a threshold (e.g., under \$500), you may configure your policy to allow POs directly without a requisition.

### 1.5.3 4.3 RFQ and Quote Comparison

**Purchasing → RFQs** — request quotes from multiple vendors for the same items.

1. Create an RFQ with line items (item, description, qty).
2. **Send** changes status to SENT.
3. Use **Record Quote** to enter each vendor's response (price per line, lead time, availability).
4. The comparison grid highlights the lowest price per line in green.
5. **Select** the winning vendor's quote.
6. **Convert to PO** creates the purchase order at the quoted prices.

**When to use RFQs:** Use RFQs for any purchase where you have multiple potential vendors, the amount is significant, or you do not have an existing contract price. The comparison grid makes the decision defensible and documented. For routine reorders of items with a Vendor Catalog price (\$4.10), you can skip the RFQ and go directly to a PO.

### 1.5.4 4.4 Purchase Orders

**Purchasing → Orders** — the core procurement transaction.

Creating a PO is similar to SO creation: choose vendor, dates, period, terms, warehouse, and add lines (item, qty, unit cost). The PO starts in OPEN status.

#### 1.5.4.1 PO Statuses

| Status    | Meaning               | What to do                               |
|-----------|-----------------------|------------------------------------------|
| OPEN      | No lines received yet | Wait for delivery; expedite if overdue   |
| PARTIAL   | Some lines received   | Receive remaining lines when they arrive |
| RECEIVED  | All lines received    | Click Bill to create the AP bill         |
| BILLED    | AP bill generated     | Pay the bill in AP → Payments            |
| CANCELLED | Voided before receipt | No further action needed                 |

#### 1.5.4.2 Receiving Goods

Click **Receive** on an OPEN or PARTIAL PO. Enter receipt number, receipt date, period, and warehouse (+ optional bin). The system: - Creates RECEIPT inventory transactions (increases on-hand) - Pushes a FIFO cost layer for each item received - Updates line statuses; sets header to PARTIAL or RECEIVED

**After receiving:** The goods are now in inventory. The on-hand count on Items list will show the increase. The vendor's bill has not yet been recorded — that happens when you click **Bill**. Do not click Bill until you have the vendor's invoice in hand and have matched it against what you actually received (three-way match).

For drop-ship POs (ship\_to = customer address), receiving does not touch inventory — it marks the linked SO lines SHIPPED instead.

### 1.5.4.3 Creating an AP Bill from a PO

Click **Bill** on a RECEIVED or PARTIAL PO. Enter bill number, bill date, due date, period, and terms. The system creates an AP bill from the received quantities. Three-way matching is tracked (PO qty vs received qty vs billed qty) and visible under **Purchasing → 3-Way Match**.

**Three-way match procedure:** Before clicking Bill, verify: 1. **PO quantity** matches what you ordered (PO line qty) 2. **Received quantity** matches what actually arrived (receipt qty) 3. **Vendor invoice quantity and price** match the PO

The **3-Way Match** report (Purchasing → 3-Way Match) flags discrepancies. Do not pay a bill with an over-billed quantity or a price higher than the PO without a manager's approval and a PO amendment.

## 1.5.5 4.5 AP Bills

**AP → Bills** — bills can be created standalone or from a PO (§4.4).

On save, the system posts GL: DR Expense / CR AP. For foreign-currency bills the USD equivalent is computed using the exchange rate at bill date.

**Void** — OPEN bills only. GL reversal posted.

**Standalone bills** (not from a PO) are appropriate for utilities, rent, subscriptions, and professional services — expenses that do not go through the PO/receiving process. For goods that were purchased and received, always create the bill from the PO to preserve three-way match tracking.

## 1.5.6 4.6 AP Payments

**AP → Payments** — record a payment to a vendor: 1. Enter payment number, vendor, date, amount, method (CHECK, ACH, WIRE), and period. 2. Click **Apply** to match it against open bills. GL: DR AP / CR Cash. FX gain/loss entries are posted for foreign-currency bills.

**Debit Memos (AP → Debit Memos)** — record a vendor credit (return or price adjustment). Posts GL: DR AP / CR Expense. Apply to open bills to reduce their balance.

**How the check run works in practice:** 1. On the scheduled check run day (weekly or bi-weekly), go to **AP → Aging**. 2. Review all bills coming due in the next 7-14 days. 3. For each vendor with due bills, create a payment in **AP → Payments** and apply it to those bills. 4. Print checks or initiate ACH transfers outside the system. 5. After the bank confirms the checks cleared, match them in **Bank Reconciliation**.

## 1.5.7 4.7 AP Aging and Vendor Statements

**AP → Aging** — open balances bucketed by days past due.

**Read the aging weekly.** A past-due AP balance means you are damaging your relationship with that vendor and may be incurring late fees. A bill in the 91+ column that you do not recognize may indicate a receiving dispute — pull the three-way match report for that vendor to investigate.

**AP → Vendor Statement** — shows bills and payments in a date range. Print as PDF or view on screen.



### 1.5.8 4.8 1099 Report

**AP → 1099 Report** — lists vendors with `is_1099 = true` and total payments in the selected year. Filter by minimum amount (default \$600). Shows a  badge for vendors missing a Tax ID.

**Year-end 1099 preparation:** In January, run this report for the prior calendar year. For each vendor shown with a  badge, contact them to obtain their W-9 (Tax ID). 1099-NEC forms must be issued by January 31. See §10.5 (Annual Procedures) for the full year-end checklist.

### 1.5.9 4.9 Check Register and Positive Pay

**AP → Check Register** — log issued checks with check number, payee, amount, bank account. Mark as cleared or void.

**AP → Positive Pay** — download a CSV file of uncleared checks in the required bank format for fraud prevention.

### 1.5.10 4.10 Vendor Rebates

**Purchasing → Vendor Rebates** — track rebate agreements (percent or fixed-per-unit) with effective and expiry dates and a minimum purchase threshold.

1. **Compute** — scans PO receipts in a date range and calculates accrued rebate amounts. Skips receipts already accrued.
2. **Settle** — creates an AP debit memo for the total pending rebate and marks accruals settled.

**When to compute rebates:** Run the Compute function at month end (or whatever period the rebate agreement specifies). The resulting debit memo appears in **AP → Debit Memos** and can be applied against open bills to that vendor. Rebate agreements often have minimum purchase thresholds — the system enforces these automatically.

### 1.5.11 4.11 Vendor Item Catalog

**Purchasing → Vendor Catalog** — stores vendor-specific part numbers, lead times, MOQ, packing quantities, and last cost per item. Mark one vendor as “preferred” per item — the preferred vendor’s cost auto-fills on PO lines.

### 1.5.12 4.12 Blanket POs

**Purchasing → Blanket POs** — standing contracts with a vendor for a total committed quantity at a fixed price.

**Release** creates a standard PO for a chosen portion of the committed quantity. Released quantities are tracked against the blanket.

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## 1.6 5. Inventory

Inventory in OpenERP is a perpetual system: every transaction (receipt, shipment, adjustment, transfer) updates on-hand in real time. There is no end-of-day batch update.



### 1.6.1 5.1 Warehouses and Bins

**Inventory → Warehouses** — create warehouses (e.g. MAIN, EAST, OVERFLOW). Each warehouse has: - Warehouse code and name - **Bins** — aisle/rack/shelf/bin-code locations within the warehouse; managed via the **Bins** modal on the warehouse row

Warehouse and bin are selectable on PO receive, inventory adjustments, and warehouse transfers.

**Bin strategy:** Create bins to match your physical layout. Common patterns are A01-01 (aisle A, rack 1, shelf 1). Assigning items to specific bins (via the preferred bin on the item) enables directed putaway and pick instructions. If your operation does not require bin-level tracking, leave the bin field blank — warehouse-level tracking is sufficient.

### 1.6.2 5.2 On-Hand Quantity

On-hand is calculated in real time as the signed sum of all inventory transactions:

| Transaction type | Effect                       |
|------------------|------------------------------|
| RECEIPT          | +qty (PO receive)            |
| SHIPMENT         | -qty (SO ship)               |
| ADJUSTMENT       | ±qty (manual)                |
| TRANSFER_OUT     | -qty (source warehouse)      |
| TRANSFER_IN      | +qty (destination warehouse) |
| RETURN_IN        | +qty (RMA receipt)           |

### 1.6.3 5.3 Manual Adjustments

**Inventory → Items** — click **Adjust** on an item row. Enter qty (positive = increase, negative = decrease), date, period, warehouse, optional bin, and notes. The system posts a GL batch: DR Inventory / CR Inv-Adj (or reversed for negative adjustments).

**When to use manual adjustments:** Use adjustments for: - Correcting a count error discovered during a cycle count - Recording spoilage, damage, or shrinkage - Loading opening inventory balances at go-live - Adjusting for goods that were consumed internally (samples, internal use)

Always enter a meaningful note in the notes field — the Audit Log will show who made the adjustment and when, and the note explains why. Unexplained adjustments are an audit risk.

### 1.6.4 5.4 Warehouse Transfers

**Inventory → Transfers** — move stock between warehouses.

1. Create a transfer: from-warehouse, to-warehouse, optional from/to bins, and line items with qty.
2. **Post** executes the transfer: creates TRANSFER\_OUT at the source and TRANSFER\_IN at the destination.
3. **Cancel** voids an unposted transfer.

**In-transit inventory:** Once you post a transfer, the source warehouse shows reduced on-hand immediately. There is no “in-transit” status — the destination warehouse shows the increase at the same moment. If your physical transfer takes days

(inter-city shipment), you may want to record the transfer only when the goods physically arrive at the destination.

### 1.6.5 5.5 FIFO Cost Layers

Every PO receipt pushes a cost layer (qty\_received, unit\_cost, receipt\_id, lot\_id). When items are shipped via SO, the system consumes the oldest layers first (FIFO). The resulting COGS is posted to GL.

**Why FIFO matters:** FIFO means the cost on your income statement reflects the oldest inventory you hold. In an inflationary environment (where costs are rising), FIFO produces a lower COGS and higher gross profit than LIFO. This is standard for most distributors and is required under IFRS. The **Valuation (FIFO)** report shows you exactly what your inventory is worth at any point in time.

**Inventory → Valuation (FIFO)** — shows current FIFO layers by item and warehouse: qty remaining, average cost per unit, and total value. A summary endpoint returns the grand total.

### 1.6.6 5.6 Landed Costs

**Purchasing → Landed Costs** — allocate additional costs (ocean freight, customs duty, brokerage, insurance, tariff) to PO receipts.

1. Create a landed cost shipment linked to one or more PO receipts.
2. Enter cost components by type and amount.
3. For tariff costs, click **Compute from item tariff rates** to auto-calculate from the items' tariff\_rate field.
4. **Post** distributes the costs proportionally across FIFO layers (by value or by qty), increasing the unit cost per layer. GL entries are not generated at this stage — the adjusted cost flows into COGS when items are shipped.

**Why landed cost matters:** If you import goods, the unit cost on the PO is typically just the supplier's price. The true landed cost includes freight, customs duties, and broker fees — often adding 10-25% to the PO price. Without landed cost allocation, your FIFO layers understate the true cost of your inventory, and your COGS is understated. Post landed costs before you ship those items; once a FIFO layer is consumed by a shipment, the layer is gone and cannot be retrospectively adjusted.

### 1.6.7 5.7 Physical Inventory Count

**Inventory → Physical Count** — a formal full-warehouse count.

1. **Create** a count session for a warehouse. The system snapshots current on-hand as system\_qty for every active item.
2. Enter counted\_qty for each line.
3. **Post** creates ADJUSTMENT inventory transactions for variances and posts GL entries for the variance value.

**Physical count best practices:** - Schedule a full physical count once per year, typically at fiscal year end or a slow period. - Freeze all receipts and shipments during the count (lock the warehouse). - Count every item at least twice; have a second person recount any item with a significant discrepancy. - Post the count as soon as possible after completing it to minimize the gap between the snapshot and the true count. - Document and review all variances over a threshold (e.g., \$500) before posting — unexplained large variances may indicate theft or systemic errors.

### 1.6.8 5.8 Cycle Counts

**Inventory → Cycle Counts** — targeted per-item counts on a rolling schedule.

1. **Schedule** a count for an item, warehouse, and optional bin with a scheduled date.
2. **Enter Count** — update counted\_qty.
3. **Approve** — creates an ADJUSTMENT transaction and posts the variance to GL.

Filter by ABC class (A/B/C) to prioritize high-value items.

**Cycle count program:** A well-run cycle count program eliminates the need for an annual shutdown physical count. The idea: - **A items** (high value, top 70% of COGS): count monthly - **B items** (mid value, next 20%): count quarterly - **C items** (low value, bottom 10%): count semi-annually

Each week, schedule the counts due that week, have warehouse staff count and enter results, and approve any within tolerance. Investigate variances above your threshold (e.g., >2% of value). Over time, cycle counts build much more accurate inventory records than an annual physical count.

### 1.6.9 5.9 Reorder Suggestions

**Inventory → Reorder Suggestions** — lists items where on-hand is below the reorder point. Shows reorder qty, min/max qty, standard cost, and reorder value. Filter by warehouse.

**How to use reorder suggestions:** 1. Check the Reorder Suggestions screen Monday morning (or daily for fast-moving items). 2. For each item, verify the reorder point and reorder qty are still correct (demand patterns change). 3. Group items by preferred vendor (visible in the Vendor Catalog). 4. Create a PO per vendor covering all items from that vendor that need replenishment. 5. The reorder qty shown is the suggested quantity; adjust up if you have a promotion coming or down if demand has slowed.

### 1.6.10 5.10 Auto-Replenishment

**Inventory → Replenishment** — click **Run** to auto-generate draft POs for all items below reorder point, grouped by preferred vendor. Review and post the draft POs from **Purchasing → Orders**.

### 1.6.11 5.11 ABC Classification

**Inventory → ABC Classify** — ranks all items by annual COGS and assigns A (top 70%), B (next 20%), C (bottom 10%). Updates the abc\_class field on each item. Run annually or after significant volume changes to keep cycle count scheduling meaningful.

### 1.6.12 5.12 Demand Forecasting

**Inventory → Items → Forecast** modal — shows 12-month transaction history and computes 3-month, 6-month, and 12-month moving average daily demand. Includes a projected date when current on-hand will reach the reorder point at the forecast demand rate.

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## 1.7 6. Item Master

### 1.7.1 6.1 Items

**Inventory → Items** — the central item catalog.

Key fields: - **Item number** — unique; used on all documents - **Description** - **Category** — for reporting and matrix pricing - **UOM** — stocking unit of measure - **Standard cost** — used for GL posting when FIFO layers are unavailable - **List price** — base selling price before price lists apply - **Reorder point / reorder qty / min qty / max qty** — drive reorder suggestions - **ABC class** — A, B, or C (updated by ABC classification run) - **HTS code, country of origin, tariff rate** — for import cost calculations - **Is kit** — if checked, the item is exploded into BOM components on SO entry

### 1.7.2 6.2 Categories and UOM

- **Inventory → Categories** — organize items for reporting and matrix pricing
- **Inventory → UOM** — units of measure; add **Conversions** (e.g. 12 EA = 1 CASE) so items can be sold in one UOM and stocked in another

### 1.7.3 6.3 Price Lists

**Sales → Price Lists** — customer-assignable price lists with per-item unit price overrides.

- **Items** button on a price list row — add or edit item overrides (unit price, effective/expiry dates)
- **Q Breaks** button on a price list item — define quantity break tiers: min\_qty, unit\_price pairs. When a line's qty\_ordered meets a break, the break price is used.

**Pricing hierarchy (highest to lowest priority):** 1. Contract price (customer + item + date range) 2. Price list assigned to the customer (with quantity breaks) 3. Matrix pricing (customer class × item category discount) 4. Item list price

If none of these apply, the order entry user must enter the price manually. Review and update price lists at least annually — or when your cost increases — to protect margins.

### 1.7.4 6.4 Contract Pricing

**Sales → Contracts** — customer + item fixed price with an effective date range. Takes priority over all price lists. Good for long-term supply agreements.

### 1.7.5 6.5 Matrix Pricing

**Sales → Matrix Pricing** — discount rules keyed on customer class × item category. Applied when no contract or price list override exists. Example: KEY customers get 15% off FITTINGS.

### 1.7.6 6.6 Lot Tracking

Items with lot\_tracking = true require a lot number on receipt and shipment. Manage lot numbers from **Inventory → Items → Lots** modal: enter lot number, received date, and optional expiry date. Expiring lots are highlighted in the lot list.

### 1.7.7 6.7 Serial Number Tracking

Items with `serial_tracking = true` require a serial number on each unit shipped. Manage serial numbers from **Inventory** → **Items** → **Serials** modal: enter serial number, received date, optional warranty expiry.

### 1.7.8 6.8 Traceability

**Inventory** → **Traceability** — look up all inventory transactions for a given serial number or lot number. Trace from receipt PO through shipment SO.

### 1.7.9 6.9 Item Cross-References

**Inventory** → **Items** → **X-Refs** modal — store alternative part numbers:

| Type       | Example use                              |
|------------|------------------------------------------|
| CUSTOMER   | Customer's own part number for this item |
| COMPETITOR | Competitor's catalog number              |
| UPC        | Universal Product Code                   |
| MPN        | Manufacturer Part Number                 |

Cross-references feed the barcode lookup endpoint.

### 1.7.10 6.10 Item Substitutes

**Inventory** → **Items** → **Subs** modal — define substitute items with priority ranking. When a flagged-superseded item is selected on an SO, the system prompts with available substitutes.

### 1.7.11 6.11 Item Supersession

**Inventory** → **Items** → **Supersession** history — record when one item replaces another (old item → new item, reason, effective date). SO entry warns the user when a superseded item is selected.

### 1.7.12 6.12 Kitting / BOM

**Inventory** → **Kitting** — bill-of-materials for kit items. Defines component items and required quantities. When a kit item is shipped, the system pulls FIFO cost from the components.

### 1.7.13 6.13 Barcode Scanning

The endpoint `GET /items/barcode/{code}` checks `item_number` first, then cross-reference numbers (UPC, MPN, etc.). Returns the matching item. Useful for integrating mobile scan devices or the WH app.

### 1.7.14 6.14 Inbound Shipments

**Purchasing** → **Inbound Shipments** — track ocean or air freight containers with vessel, voyage, carrier, origin/destination ports, ETA, ATA, and customs status (PENDING → IN\_TRANSIT → ARRIVED → CLEARED → HELD).

Attach POs to a shipment to group expected receipts by container.

## 1.8 7. Financials

The Financials module provides the General Ledger, financial statements, bank reconciliation, multi-currency, and year-end close. For most companies, the subledgers (AR, AP, Inventory) handle the day-to-day posting automatically. The GL is where you review the cumulative result, make manual corrections, and produce financial statements for management and external reporting.

### 1.8.1 7.1 General Ledger

**GL → Accounts** — chart of accounts (see §1.4).

**GL → Fiscal Years / Periods** — set up the accounting calendar. Closing a period prevents any further posting to it. Fiscal year close (§7.7) is a separate workflow.

**GL → Journals** — manual journal entry: 1. Create a batch (batch number, description, period). 2. Add a journal entry (entry number, date, source, reference, description). 3. Add debit and credit lines. The batch will not post unless total debits = total credits. 4. **Post** the batch. Once posted, entries cannot be edited; a reversing entry is required.

**When to use manual journals:** - Depreciation not handled by the Fixed Assets module - Accruals (record an expense incurred but not yet billed) - Reversals (reverse a prior period accrual) - Reclassifications (move an amount from one account to another) - Opening balance entries at go-live

Do **not** use manual journals for: AR invoices, AR payments, AP bills, AP payments, SO shipments, or inventory adjustments. Those events post automatically. Creating a manual journal for something the system already posted will double-count the amount.

Subledger-to-GL auto-posting happens silently when AR invoices, AR payments, AP bills, AP payments, SO shipments, and inventory adjustments are saved. You generally do not need to create manual journals for those events.

### 1.8.2 7.2 Financial Statements

**GL → Reports:** - **Trial Balance** — all accounts with period debit/credit activity and ending balance. Requires a period selection. - **Income Statement** — revenue minus expense for a period or date range. - **Balance Sheet** — assets, liabilities, and equity as of a period end.

**How to read the Trial Balance as a diagnostic:** The Trial Balance should always balance (total debits = total credits). Run it at month end before closing the period. Look for: - Accounts with balances in the wrong direction (e.g., a REVENUE account with a debit balance — may indicate a void posted to the wrong period) - Unexpectedly large or small balances compared to prior months - Accounts with activity you do not recognize

Investigate anomalies before closing the period. It is much harder to correct after the period is closed.

### 1.8.3 7.3 GL Budgets

**GL → Budgets** — enter budget amounts per account per period. The **Budget vs Actual** report compares YTD actual to YTD budget.

### 1.8.4 7.4 Departments

**GL → Departments** — cost centers (e.g. SALES, WAREHOUSE, ADMIN). Journal lines can be assigned to a department for departmental P&L analysis.

### 1.8.5 7.5 Fixed Assets and Depreciation

**GL → Fixed Assets** — track capital assets with: - In-service date, useful life (months), salvage value - Linked GL asset account and accumulated depreciation account - **Depreciate** button — posts straight-line monthly depreciation entries for the selected period

**GL → Depreciation Report** — shows all assets with book value, accumulated depreciation, and remaining life.

**Monthly depreciation procedure:** At month end, go to **GL → Fixed Assets** and click **Depreciate** for each active asset. This posts the depreciation journal entry for that period. Do this before running the Income Statement so that the depreciation expense is included in the current period's result.

### 1.8.6 7.6 Recurring Journal Entries

**GL → Recurring JEs** — templates for monthly entries (rent, amortization, etc.): - Define template lines (debit/credit amounts and accounts). - Set frequency (MONTHLY, QUARTERLY, ANNUALLY) and next run date. - **Run** generates a journal batch from the template and advances the next run date.

**Common recurring entries to set up:** - Rent (DR Rent Expense / CR Prepaid Rent or CR Accounts Payable) - Insurance prepaid amortization (DR Insurance Expense / CR Prepaid Insurance) - Loan interest accrual (DR Interest Expense / CR Accrued Interest) - Deferred revenue recognition (DR Deferred Revenue / CR Revenue)

Setting these up as recurring templates ensures they are never forgotten and the amounts are consistent each month.

### 1.8.7 7.7 Year-End Close

**GL → Year-End Close** (when fiscal year is fully posted): 1. Close all open periods in the fiscal year. 2. **Run Close** — the system: - Zeros all revenue and expense account balances for the year via a closing journal entry - Credits the net income to the Retained Earnings account - Marks the fiscal year CLOSED 3. Create the new fiscal year and its periods.

**Year-end close is irreversible.** Before running it, confirm: - All AR invoices and payments for the year are entered - All AP bills and payments for the year are entered - All physical count variances are posted - All fixed asset depreciation entries are posted - All recurring journal entries have been run for December - The Trial Balance balances and all balances look correct - A backup of the database has been taken

The year-end close does not affect AR or AP balances — those carry forward as open items. Only income statement accounts (Revenue, Expense) are zeroed.

### 1.8.8 7.8 Bank Reconciliation

**Financials → Bank Reconciliation:** 1. Select a bank account and date range. 2. **Add Statement Lines** — enter or import bank statement transactions (deposits positive, payments negative). 3. **Match** — link each statement line to an AR receipt (deposit) or AP payment from



the system. 4. Summary tiles show: statement total, cleared total, uncleared total, outstanding deposits, outstanding payments. 5. The reconciliation is complete when uncleared items equal the difference between the bank statement balance and the GL cash account balance.

**Bank reconciliation procedure (monthly):** 1. Obtain the bank statement (paper, PDF, or CSV download from your bank's portal). 2. Enter all statement lines into the system using **Add Statement Lines**, or import them. 3. Match each statement line to a payment or deposit in the system using the **Match** button. 4. Common unmatched items: - **Outstanding checks** — payments created in the system but not yet cleared at the bank - **Deposits in transit** — AR receipts recorded in the system but not yet on the statement - **Bank charges** — service fees on the statement with no corresponding system entry; create a manual journal (DR Bank Charges / CR Cash) 5. The reconciliation is complete when: **Bank Statement Balance + Outstanding Deposits – Outstanding Checks = GL Cash Account Balance**. 6. Reconcile monthly, ideally within 5 business days of receiving the statement.

### 1.8.9 7.9 Multi-Currency

**Admin → Currencies** — activate currencies and set exchange rates (spot rates by date).

Foreign-currency AP bills store the foreign amount and the exchange rate at bill date. When the bill is paid at a different rate, the system computes and posts FX gain/loss to GL automatically.

**GL → FX Revaluation** — revalue all open foreign-currency AR/AP balances at the current exchange rate and post unrealized gain/loss.

**Multi-currency procedure:** At month end, if you have open foreign-currency AR or AP balances, run **FX Revaluation** to record unrealized gains and losses at the current month-end rate. This is required for accurate financial statements under US GAAP and IFRS. The revaluation entries reverse in the following period, so only the realized gain/loss (from payment) remains permanently.

### 1.8.10 7.10 Cash Flow Statement

**GL → Cash Flow** — indirect method: starts from net income, adds back depreciation, adjusts for AR/AP/inventory working capital changes, and includes asset purchases. Reads from GL account balances by subtype.

### 1.8.11 7.11 Cash Application Batch

**AR → Batch Apply** — apply multiple payments at once. The system matches each payment to the customer's oldest open invoices first (oldest-invoice-first auto-apply). Returns a per-payment result showing applied amounts and any unapplied balance.

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## 1.9 8. Administration

### 1.9.1 8.1 Users and Groups

**Admin → Users:** - Create users with a username, password (minimum 6 characters), full name, and group assignment. - Deactivate users to block login without deleting records. - **Change Password** — bumps token\_version to immediately invalidate all active sessions for



that user. - **Force Logout** — invalidates all active tokens for the user without changing the password.

**Admin → Groups:** - Groups bundle permissions. Every user belongs to exactly one group. - Create groups with a name and description, then check the permissions to grant. - Changing a group's permissions takes effect on the user's next request (token is re-validated per request).

**Recommended group structure:**

| Group     | Who                 | Typical permissions   |
|-----------|---------------------|-----------------------|
| ADMIN     | Controller, IT      | All permissions       |
| AR_STAFF  | AR clerk            | AR module only        |
| AP_STAFF  | AP clerk            | AP module only        |
| WAREHOUSE | Receiving, shipping | Inventory, PO receive |
| SALES     | Inside sales        | SO, quotes, customers |
| READ_ONLY | Managers, auditors  | View only             |

Separate AR and AP access so that no single employee can both record a payment receipt AND apply it to an invoice without a second person reviewing. This is a basic internal control.

## 1.9.2 8.2 Permissions Reference

| Permission key       | What it gates                                       |
|----------------------|-----------------------------------------------------|
| system.users.manage  | Admin → Users and Admin → Groups                    |
| system.config.manage | Admin → Company Settings, Audit Log                 |
| gl.reports.view      | GL Trial Balance, Income Statement, Balance Sheet   |
| ar.invoices.view     | AR → Recurring Invoice Templates                    |
| (all other modules)  | require_auth() only — any logged-in user can access |

Note: most module routes require only a valid login (not a specific permission). Per-route permission enforcement is an active roadmap item (D4).

## 1.9.3 8.3 Audit Log

**Admin → Audit Log** — every data-changing API call logs: table name, record ID, action (CREATE, UPDATE, DELETE, POST, APPLY, SHIP, etc.), changed-by username, and timestamp. Filter by table name, record ID, user, or date range.

**Using the audit log for investigations:** If a balance is unexpectedly wrong, filter the audit log by the affected record ID and review the sequence of actions. You can see exactly who changed what and when. The audit log is append-only and cannot be edited.

## 1.9.4 8.4 Payment Terms

**Admin → Payment Terms** — manage NET30, NET60, 2/10-NET30, etc. Payment terms drive due-date calculation on invoices and bills.

### 1.9.5 8.5 Tax Codes

**Admin → Tax Codes** — simple flat-rate tax codes (e.g. TX-STATE at 6.25%). Assigned to items; defaulted onto invoice lines.

### 1.9.6 8.6 Sales Tax (US Multi-Jurisdiction)

**Admin → Sales Tax** — jurisdiction-based rates: - Create jurisdictions at state, county, and city level with state/county/city rate components. - Assign **exemption certificates** to customers (expiry date, certificate number) — exempted customers have 0% tax applied. - On an AR invoice, the **Sales Tax** panel computes the blended rate for each line's jurisdiction and applies it via **Apply to Lines**.

### 1.9.7 8.7 Customer Classes

**Admin → Customer Classes** — codes used for matrix pricing and reporting (e.g. KEY, STANDARD, PROSPECT). Assign to customers on the customer form.

### 1.9.8 8.8 Return Reason Codes

**Admin → Return Reasons** — standardized codes for RMA lines (e.g. DEFECTIVE, WRONG\_ITEM, DAMAGED). Appear in the RMA line form dropdown.

### 1.9.9 8.9 Carriers and Ship Methods

**Admin → Carriers** — carrier master (UPS, FedEx, etc.). Linked to ship methods.

**Admin → Ship Methods** — specific service levels per carrier (UPS GROUND, FEDEX 2DAY). Assignable on SOs and POs.

### 1.9.10 8.10 SMTP and Email Log

Configure SMTP under **Admin → Company Settings** (smtp\_host, smtp\_port, smtp\_encryption, smtp\_user, smtp\_pass, smtp\_from). Use **Test Send** to verify delivery.

**Admin → Email Log** — every email sent by the system (invoice, SO, PO) is recorded with status, recipient, subject, and timestamp.

### 1.9.11 8.11 Scheduled Reports

**Admin → Scheduled Reports** — configure reports to run automatically and email results: - Report type (AR Aging, AP Aging, Trial Balance, Backorders, Commissions) - Frequency (DAILY, WEEKLY, MONTHLY) - Next run date - Recipients (comma-separated email addresses)

Run the scheduler manually:

```
C:\php\php.exe -c C:\php\php_sapads.ini F:\OpenERP\tools\run_scheduled_reports.php
```

Or configure Windows Task Scheduler to run this script at your desired interval (e.g., daily at 7:00 AM).

### 1.9.12 8.12 CSV Import

**Admin → Import Tool** — bulk-load master data:

| Import type | Natural key (for upsert) | Notes                       |
|-------------|--------------------------|-----------------------------|
| Customers   | customer_number          | Upserts: creates or updates |
| Vendors     | vendor_number            | Upserts                     |
| Items       | item_number              | Upserts                     |
| GL Balances | account_id + period_id   | Upserts opening balances    |

Download the column template from the Import Tool page. Required columns are documented in the template header row.

### 1.9.13 8.13 CSV Export

Most list screens have an **Export CSV** button that downloads the current filtered/paginated view as a CSV file. Available on: AR Invoices, AR Customers, AP Vendors, AP Bills, Sales Orders, and the Inventory Valuation report.

## 1.10 9. Integration

### 1.10.1 9.1 EDI — Inbound Purchase Orders (850)

**Integration → EDI Partners** — configure trading partners (ISA sender/receiver IDs, partner name, direction).

To import an EDI 850 (purchase order from a customer treated as a vendor PO to you):

POST /edi/inbound/850

Body: { "partner\_id": "<edi\_partner\_id>", "x12\_content": "<full X12 string>" }

The parser auto-detects element and segment separators from the ISA fixed positions, then creates a PO and lines. Results are logged to edi\_transactions.

### 1.10.2 9.2 EDI — Outbound ASN (856) and Invoice (810)

GET /edi/outbound/856/{so\_id} → X12 856 ASN file download

GET /edi/outbound/810/{inv\_id} → X12 810 invoice file download

These return the X12 flat-file as a text response. Use your EDI VAN or AS2 adapter to transmit to the trading partner.

### 1.10.3 9.3 E-Commerce Webhook

**Integration → E-Commerce Orders** — shows orders imported via webhook.

To push an order from your e-commerce platform:

POST /ecommerce/webhook

```
Body: {
  "source": "shopify",
  "external_order_id": "12345",
  "customer_email": "...",
  "ship_to": { "name": "...", "address1": "...", "city": "...", "state": "...", "zip": "..." },
  "lines": [{ "sku": "ITEM-001", "qty": 2, "unit_price": 15.00 }]
}
```

The system finds or creates the customer by email, creates an SO with the lines, and deduplicates by source + external\_order\_id — safe to replay.

### 1.10.4 9.4 Recurring POs

**Purchasing → Recurring POs** — templates for regularly-ordered items (weekly, monthly). Configure a template with vendor, lines, and frequency. **Run** generates a PO from the template and advances the next run date.

Automate with:

```
C:\php\php.exe -c C:\php\php_sapads.ini F:\OpenERP\tools\run_po_recurring.php
```

---

## 1.11 10. Management Operating Procedures

This section describes the recommended business routines for running the system efficiently. These are operating procedures, not software instructions — they describe *when* and *why* to perform tasks, not just *how*.

### 1.11.1 10.1 Daily Routines

#### 1.11.1.1 Finance / Controller

- **Check the Dashboard.** Review cash, open AR, open AP, and overdue AR tiles. Any sharp movements overnight indicate transactions that need review.
- **Process AR receipts.** Enter payments received by mail or wire from the prior day into **AR → Payments** and apply them to open invoices.
- **Process AP bills.** Enter vendor invoices received by mail or email into **AP → Bills**. Code each bill to the correct period and GL account.
- **Review the Audit Log** (if anomalies are suspected). Filter by yesterday's date to see all transactions.

#### 1.11.1.2 Warehouse / Receiving

- **Receive open POs.** For each delivery that arrives, locate the PO in **Purchasing → Orders** and click **Receive**. Enter the actual quantities received. Print the receipt for the packing file.
- **Check reorder suggestions.** After receiving, review **Inventory → Reorder Suggestions** to see if any items have dropped below reorder point. Notify purchasing.

#### 1.11.1.3 Sales / Customer Service

- **Check the back-order report.** Review **Sales → Back-Orders** for any lines with a past ship date. Contact customers about delays and update promise dates.
- **Process new orders.** Enter customer orders promptly — the ATP check gives the customer real-time availability.

### 1.11.2 10.2 Weekly Routines

#### 1.11.2.1 AR Collections (every Monday, or first business day of the week)

1. Run **AR → Aging**. Sort by the 61-90 and 91+ columns descending.

2. For every customer with a balance in the 61–90 column: send a statement (AR → Statements → Email) or make a phone call. Log the contact in Collections.
3. For every customer with a balance in the 91+ column: escalate — contact the customer directly, involve the sales rep, and consider placing the account on credit hold (AR → Customers → Hold).
4. For customers on hold, the system blocks new orders automatically. Release the hold (AR → Customers → Release) only when payment is received or a payment plan is agreed.
5. Discuss write-off candidates with management. Write-offs over a threshold (e.g., \$1,000) should require controller approval.

#### 1.11.2.2 AP Check Run (typically Wednesday or Thursday)

1. Run **AP → Aging**. Identify all bills due within the next 7 days.
2. Add bills that offer an early-pay discount (e.g., 2/10-NET30) if the discount is worth more than your cost of funds.
3. Create payments in **AP → Payments** for each vendor.
4. Print checks or initiate ACH/wire transfers through your bank.
5. After the bank processes the payments, match them in **Bank Reconciliation** when the statement updates.

#### 1.11.2.3 Purchasing Review

1. Review open POs past their expected delivery date (**Purchasing → Orders**, filter OPEN/PARTIAL, sort by need-by date).
2. Contact vendors on overdue deliveries and update the expected date.
3. Review **Inventory → Reorder Suggestions** for any new shortfalls.
4. Process any RFQs that have received all vendor responses — compare and convert the winner to a PO.

#### 1.11.2.4 Sales Commission Review

1. Run **Sales → Commissions** for the current month-to-date.
2. Share with sales management to track rep performance against quota.
3. Commissions accrue on invoiced SOs only (not on orders or shipments) — remind reps to push customers to accept invoices promptly.

### 1.11.3 10.3 Monthly Period-End Checklist

Perform these steps in order, on or before the last business day of each month. Do not close the period until all steps are complete.

**Step 1 — Complete all transactions for the period.** - Ensure all AR invoices for the month are entered and emailed. - Ensure all AP bills received during the month are entered. - Ensure all PO receipts for the month have been received in the system. - Ensure all cash receipts received during the month are applied. - Ensure all AP payments issued during the month are recorded.

**Step 2 — Inventory.** - Post any pending cycle count approvals. - Post any pending physical count sessions. - Run landed cost allocation for any containers that arrived during the month (**Purchasing → Landed Costs**). - Review **Inventory → Valuation (FIFO)** — does the total inventory value look reasonable?

**Step 3 — GL Adjustments.** - Run all recurring journal entries for the month (**GL → Recurring JEs**). - Run depreciation for all fixed assets (**GL → Fixed Assets → Depreciate** for each

asset). - Post any accruals required by your accountant (accrued wages, accrued expenses, deferred revenue). - If you have foreign currency balances, run **GL → FX Revaluation**.

**Step 4 — Reconcile.** - Reconcile the bank account in **Financials → Bank Reconciliation** for the month just ended. - Confirm the GL Cash balance matches: Bank Statement Balance + Outstanding Deposits – Outstanding Checks. - Reconcile the AR subledger to the GL: the sum of open AR invoice balances should equal the GL AR account balance. - Reconcile the AP subledger to the GL: the sum of open AP bill balances should equal the GL AP account balance.

**Step 5 — Review financial statements.** - Run the **Trial Balance** for the period. Confirm it balances. Investigate any unusual account balances. - Run the **Income Statement**. Compare to prior month and prior year same month. Explain significant variances. - Run the **Balance Sheet**. Confirm AR, AP, and Inventory totals match the reconciliations in Step 4.

**Step 6 — Close the period.** - Once satisfied that the period is correct, go to **GL → Periods** and click **Close** for the month. - A closed period cannot receive new transactions. If a correction is needed after closing, it must be posted in the next open period with a reversing entry pair.

**Step 7 — Distribute reports.** - Send the Income Statement and Balance Sheet to management. - Send the AR Aging to the collections team. - Send the AP Aging to the controller / CFO. - Send the Sales Commission report to the sales manager. - Send the Inventory Valuation summary to the operations manager.

#### 1.11.4 10.4 Quarterly Procedures

**Sales Tax Filing:** - Run **Admin → Sales Tax** reports for the quarter. - Calculate tax collected by jurisdiction and remit to the applicable state/local tax authorities. - File sales tax returns as required by each jurisdiction.

**Inventory ABC Review:** - Run **Inventory → ABC Classify** to re-rank items by actual COGS for the quarter. - Adjust cycle count frequencies based on the new ABC classification. - Review reorder points for A items — these should be rechecked against current demand patterns.

**Vendor Performance Review:** - Pull **Purchasing → 3-Way Match** with the Exception filter to find vendors with chronic over-billing or over-receiving issues. - Review **Purchasing → Inbound Shipments** for vendors with chronic late deliveries. - Use this data in vendor negotiations.

**Budget vs Actual:** - Run **GL → Budgets → Budget vs Actual** for the quarter. - Identify accounts that are significantly over or under budget. - Adjust forecasts for the remainder of the year and communicate to department heads.

#### 1.11.5 10.5 Annual Procedures

##### 1.11.5.1 Year-End Financial Close

1. **Complete all December transactions** using the Monthly Period-End Checklist (§10.3).
2. **Physical inventory count** — schedule and complete a full physical count in the last week of December. Post all variances.
3. **Depreciation** — run December depreciation for all fixed assets.
4. **Accruals** — consult your accountant for year-end accruals:
  - Accrued bonus/vacation
  - Prepaid expenses to amortize
  - Deferred revenue
5. **Close all 12 periods** in the fiscal year.

6. **Run Year-End Close** — go to **GL → Year-End Close**, review the summary, and click Run Close. This zeros income statement accounts and credits net income to Retained Earnings.
7. **Create the new fiscal year** and all 12 periods.
8. **Verify opening balances** — run the Balance Sheet for the new year's first period. All balance sheet accounts should show the same closing balance as December 31.

#### 1.11.5.2 1099 Preparation (US)

1. In January, run **AP → 1099 Report** for the prior calendar year with a minimum of \$600.
2. For each vendor with a □ badge (missing Tax ID), contact the vendor and obtain a completed W-9.
3. Enter the Tax ID on the vendor record in **AP → Vendors**.
4. Print or e-file 1099-NEC forms. Due to recipients: **January 31**. Due to IRS: **January 31** (electronic) or **February 28** (paper).

#### 1.11.5.3 Price List and Contract Review

1. Run a margin analysis by item category — pull Sales data vs. FIFO cost from the Inventory Valuation report.
2. Identify items where your cost has risen but list prices have not been updated.
3. Update price lists and quantity breaks in **Sales → Price Lists**.
4. Review and renew (or renegotiate) customer contracts in **Sales → Contracts**.

#### 1.11.5.4 Sales Rep Commission Reconciliation

- Pull the full-year **Sales → Commissions** report.
- Reconcile to any commission statements you have paid manually during the year.
- This is the basis for annual commission W-2 or 1099 reporting.

### 1.11.6 10.6 Exception Handling

#### 1.11.6.1 Customer Disputes a Bill

1. Review the original SO, ship confirmation, and AR invoice — confirm quantities and pricing.
2. If the customer is correct: issue a **Credit Memo** (AR → Credit Memos) for the disputed amount and apply it to the invoice.
3. If the customer is wrong: send a copy of the original documentation and the invoice PDF. Log the dispute in the Collections notes.
4. Never adjust an invoice after it has been sent — create a credit memo instead. Changing a posted invoice creates audit issues.

#### 1.11.6.2 Vendor Overcharges on a Bill

1. Do not pay the overcharged amount.
2. Contact the vendor and request a corrected invoice or a credit memo.
3. If the vendor issues a credit memo, enter it as an **AP Debit Memo** (AP → Debit Memos) and apply it against the bill.
4. If the vendor refuses and the amount is material, void the original bill, enter the correct amount as a new bill, and document the situation in the notes.



### 1.11.6.3 Inventory Discrepancy

1. Identify the item, warehouse, and bin.
2. Count the actual quantity on hand.
3. Review recent inventory transactions in the Audit Log — look for unauthorized adjustments or receiving errors.
4. If the count is correct and the system is wrong: post an adjustment in **Inventory → Items → Adjust** with a note explaining the investigation.
5. If the discrepancy is large or unexplained: investigate before adjusting — unexplained shrinkage may indicate theft.

### 1.11.6.4 Bank NSF (Returned Check)

1. The bank will debit your account for the returned check amount.
2. In the system, void the original AR payment (**AR → Payments → Void**). This reverses the application and restores the invoice balance.
3. Post a manual journal: DR Bank Charges / CR Cash for the NSF fee the bank charges you.
4. Contact the customer immediately. Place their account on credit hold pending resolution.
5. Re-invoice the customer for the NSF fee if your policy allows.

## 1.12 11. Reference

### 1.12.1 11.1 Document Status Codes

**Sales Orders and Quotes:** DRAFT → SENT → ACCEPTED / EXPIRED / CANCELLED (quotes) OPEN → PARTIAL → SHIPPED → INVOICED / CANCELLED (orders)

**Purchase Orders and Blanket POs:** OPEN → PARTIAL → RECEIVED → BILLED / CANCELLED

**Purchase Requisitions:** DRAFT → PENDING → APPROVED / REJECTED → CONVERTED / CANCELLED

**RFQs:** DRAFT → SENT → CLOSED / CONVERTED

**RMAs and Vendor RMAs:** PENDING → APPROVED → RECEIVED → CREDITED / CLOSED / CANCELLED

**AP Bills / AR Invoices:** OPEN → PARTIAL → PAID / VOID

**GL Batches:** OPEN → POSTED

**Inbound Shipments (containers):** PENDING → IN\_TRANSIT → ARRIVED → CLEARED / HELD

### 1.12.2 11.2 PDF Generation

Every major document has a **PDF** button or link:

| Document                   | Endpoint                             |
|----------------------------|--------------------------------------|
| AR Invoice                 | GET /ar/invoices/{id}/pdf            |
| Customer Statement         | GET /ar/customers/{id}/statement/pdf |
| Sales Order / Packing List | GET /so/{id}/pdf                     |
| Purchase Order             | GET /po/{id}/pdf                     |
| Certificate of Conformance | GET /certs/{id}/pdf                  |



PDFs embed the company logo, configurable header subtitle, footer note, and per-document terms. Customize these under **Admin → Company Settings → Document Templates**.

### 1.12.3 11.3 GL Auto-Posting Summary

The following events automatically post a GL batch:

| Event                           | Debit                | Credit                             |
|---------------------------------|----------------------|------------------------------------|
| AR Invoice created              | AR                   | Revenue (+ Tax Payable if tax > 0) |
| AR Payment applied              | Cash                 | AR                                 |
| AP Bill created                 | Expense              | AP                                 |
| AP Payment applied              | AP                   | Cash                               |
| SO Shipment                     | COGS                 | Inventory                          |
| Inventory Adjustment (positive) | Inventory            | Inv-Adj                            |
| Inventory Adjustment (negative) | Inv-Adj              | Inventory                          |
| AR Credit Memo created          | Revenue              | AR                                 |
| AP Debit Memo created           | AP                   | Expense                            |
| AR Invoice void                 | Revenue              | AR (reversal)                      |
| AR Payment void                 | AR                   | Cash (reversal)                    |
| AP Bill void                    | AP                   | Expense (reversal)                 |
| AP Payment void                 | Cash                 | AP (reversal)                      |
| Deposit applied to invoice      | Deposits liability   | AR                                 |
| Year-end close                  | Revenue / Expense    | Retained Earnings                  |
| Monthly depreciation            | Depreciation Expense | Accum. Depreciation                |
| FX gain on AP payment           | AP                   | FX Gain                            |
| FX loss on AP payment           | FX Loss              | AP                                 |

Account resolution order: company\_config key → first active non-header account with matching subtype → error logged, transaction still committed.

### 1.12.4 11.4 API Base URL

The REST API is at `http://<server>:8081`. All endpoints require Authorization: Bearer <token> except `POST /auth/login`.

Common patterns: - `GET /ar/customers?page=1&per_page=50&search=acme` — paginated list with search - `POST /ar/customers` — create - `PUT /ar/customers/{id}` — update - `GET /ar/customers/{id}` — single record

Paginated responses include total, page, per\_page, and data fields.

### 1.12.5 11.5 Diagnostic Tools

Located in the tools/ directory (run with `C:\php\php.exe -c C:\php\php_sapads.ini tools\<script>.php`):

| Script        | Purpose                                                |
|---------------|--------------------------------------------------------|
| install.php   | Interactive first-run installer / configuration wizard |
| create_dd.php | Create or refresh the SAP ACE data dictionary          |
| seed.php      | Seed payment terms, tax codes, and chart of accounts   |

| Script                    | Purpose                                               |
|---------------------------|-------------------------------------------------------|
| seed_auth.php             | Seed permissions, groups, and admin user              |
| seed_demo.php             | Load a full demo dataset (Meridian Industrial Supply) |
| setup_users.php           | Create the openerp_app ADS user and lock down the DD  |
| smoke_test.php            | End-to-end HTTP smoke test (42 checks)                |
| rbac_test.php             | RBAC validation (32 checks)                           |
| run_scheduled_reports.php | Execute due scheduled reports and email results       |

### 1.12.6 11.6 Primary Key Format

All record IDs are 13-character strings in the format MMYNnnnnnnnn: - MM — 2-character table mnemonic (e.g. AR for invoices, SO for sales orders) - YY — 2-digit year - nnnnnnnn — 8-digit zero-padded sequence number

Example: AR26-00000042 is the 42nd AR invoice record created in 2026. IDs are generated by the NewSeqKey() database function.

### 1.12.7 11.7 Session and Security Notes

- Tokens expire after **8 hours**. The frontend redirects to login automatically.
- **Logout** (POST /auth/logout) immediately invalidates all sessions for that user — other browser tabs will be signed out on their next API call.
- **Password change** and **admin force-logout** also invalidate sessions.
- All traffic should run over HTTPS in production. Configure a reverse proxy (nginx, IIS ARR) to terminate TLS in front of port 8081.
- The data dictionary is locked (ADS\_DD\_LOG\_IN\_REQUIRED) so anonymous connections are rejected at the database level. The backend connects as the openerp\_app ADS user.

### 1.12.8 11.8 Internal Controls Checklist

These are basic controls a controller or auditor would expect to find. Review periodically:

| Control                                               | How to verify in OpenERP                                                |
|-------------------------------------------------------|-------------------------------------------------------------------------|
| No single employee can create and apply an AR payment | AR staff should have AR access; only the controller approves write-offs |
| All AP bills have a matching PO and receipt           | Run 3-Way Match report monthly; investigate exceptions                  |
| Inventory adjustments are documented                  | Audit Log → filter by table inventory_transactions and action ADJUST    |
| Terminated employees are immediately blocked          | Admin → Users → Deactivate on last day of employment                    |
| Bank accounts are reconciled monthly                  | Financials → Bank Reconciliation; confirm cleared dates match statement |
| Period is closed after month-end review               | GL → Periods; all prior periods should show CLOSED status               |

| Control                                             | How to verify in OpenERP                                      |
|-----------------------------------------------------|---------------------------------------------------------------|
| Physical count performed at least annually          | Physical Count sessions; post within 5 business days of count |
| Financial statements reviewed by management monthly | GL → Reports; Trail Balance, Income Statement, Balance Sheet  |